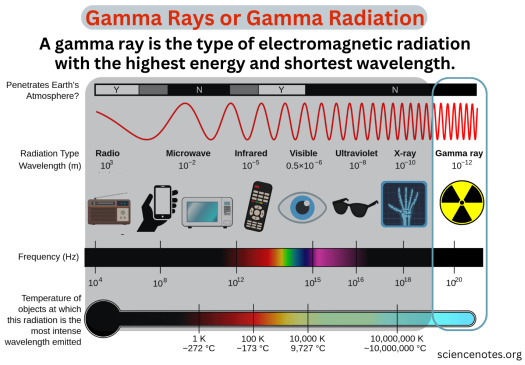
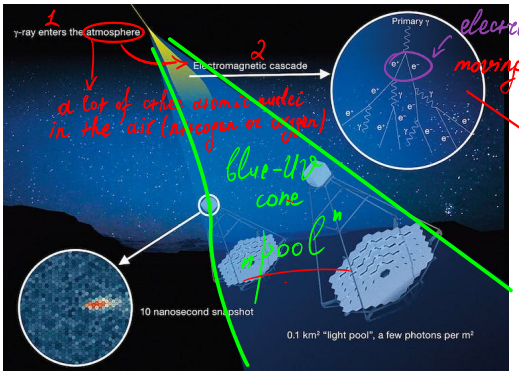
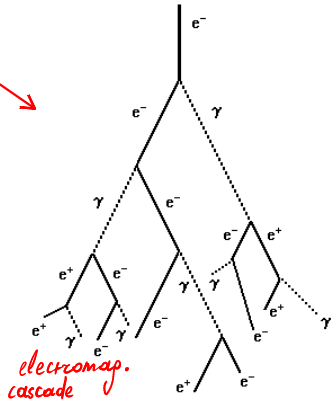
Explanation of the MAGIC Telescope.

Our Job is to detect gamma γ rays, highest frequency and energy of a form of light.
from extreme astrophysical objects.

We have a problem: gamma particles don't! reach the ground. It hits Earth's atmosphere

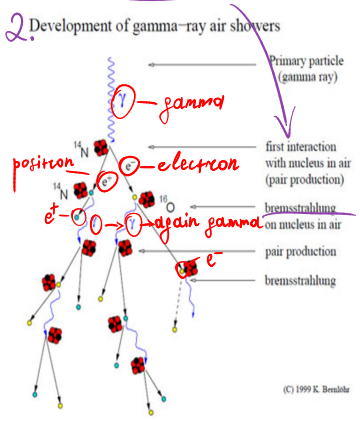






1. γ enters the atmosphere, it passes near the strong electric field of an atomic nucleus of an oxygen or nitrogen in the air → photon disappears → electron positron pair

2. Development of gamma-ray air showers



Interaction is electromagnetic, means that the shower stays narrow and well-organized, no random sideways kicks. Like a focused beam.
Only photons, electrons, positrons.

Electromag. cascade

3. v_{e^-,e^+}^{shower} in air > c air → when that speed of e^+, e^- is reached it creates a cone of blue-UV light (Cherenkov radiation) possible
By the time cone reaches the ground it creates a roughly circular "pool."

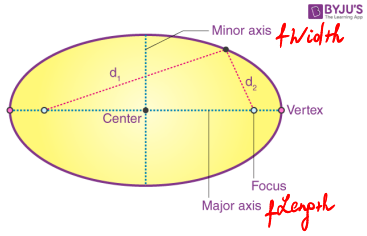
4. Finally, takes a 10^{-8} s. snapshot.

For Our ML Task: Hadrons look similar. For every such γ-ray about 10K hadrons also hit the atmosphere. But! hadronic showers develop differently.

hadrons → pions (π^+, π^-, π^0) → decays further into γ-rays → creating their own messy electromag. sub-cascades.
hadrons → kaons → create more hadrons.
hadrons → other hadrons.

Here we get the Hillas params.:

each camera image → into parametrizing as an ellipse using PCA. fLength – major axis, fWidth – minor axis of the ellipse.



fSize – $\log_{10}(1\{if=photon\})$

The rest will be in chat.